

Test Case Design Document

OldPhonePad Coding Challenge — xUnit / .NET 8

Project: OldPhonePad Coding Challenge
Component: OldPhonePad(string input) method
Version: 1.0
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Framework: xUnit .NET 8

1. Scope

This document covers 27 test cases for the OldPhonePad method, which simulates an old mobile phone keypad. Repeated key presses cycle through letters; a space separates consecutive presses of the same key; * acts as backspace; and # sends (terminates) the input.

2. Test Case Table

Test ID	Category	Description	Preconditions	Input	Expected Output	Test Type
TC-01	PDF Examples	Single letter E from repeated 3 presses	Method available; valid input string	33#	E	Positive
TC-02	PDF Examples	Backspace removes last char leaving B	Method available; valid input string	227*#	B	Positive
TC-03	PDF Examples	Multi-key sequence spells HELLO	Method available; valid input string	4433555 555666#	HELLO	Positive
TC-04	PDF Examples	Complex sequence with backspace spells TURING	Method available; valid input string	8 88777444666*664#	TURING	Positive
TC-05	Single Key Press	First letter of key 2 (A)	Method available; valid input string	2#	A	Positive
TC-06	Single Key Press	First letter of key 3 (D)	Method available; valid input string	3#	D	Positive
TC-07	Single Key Press	First letter of key 9 (W)	Method available; valid input string	9#	W	Positive

Test ID	Category	Description	Preconditions	Input	Expected Output	Test Type
TC-08	Space Separator	Space between same key presses yields AA	Method available; valid input string	2 2#	AA	Positive
TC-09	Space Separator	Key 0 produces space between A and B	Method available; valid input string	2022#	A B	Positive
TC-10	Cycling Wrap-Around	4 presses on 7 wraps to S	Method available; valid input string	7777#	S	Positive
TC-11	Cycling Wrap-Around	5 presses on 7 wraps back to P	Method available; valid input string	77777#	P	Positive
TC-12	Cycling Wrap-Around	4 presses on 2 wraps to A	Method available; valid input string	2222#	A	Positive
TC-13	Backspace	Backspace removes second A leaving B	Method available; valid input string	227*#	B	Positive
TC-14	Backspace	Backspace on empty buffer yields empty string	Method available; valid input string	*#	""	Edge Case
TC-15	Backspace	Two backspaces erase both chars of CC	Method available; valid input string	222**#	""	Positive
TC-16	Backspace	Multiple backspaces erase all typed chars	Method available; valid input string	222 33***#	""	Positive
TC-17	Backspace	Backspace removes E leaving BA	Method available; valid input string	22 33*2#	BA	Positive
TC-18	Backspace	Backspace after 0-space erases A then re-types A	Method available; valid input string	20*2#	AA	Positive
TC-19	Backspace Until Empty	Excess backspaces safely clear buffer to A	Method available; valid input string	22 2**2#	A	Edge Case
TC-20	Key 1 Special Chars	Key 1 first press gives &	Method available; valid input string	1#	&	Positive
TC-21	Key 1 Special Chars	Key 1 second press gives '	Method available; valid input string	11#	'	Positive
TC-22	Key 1 Special Chars	Key 1 third press gives (	Method available; valid input string	111#	(	Positive
TC-23	All Keys	One press of every key in order	Method available; valid input string	2345678901#	ADGJMPTW &	Positive

Test ID	Category	Description	Preconditions	Input	Expected Output	Test Type
TC-24	Defensive / Edge	Null input throws ArgumentNullException	Method available	null	ArgumentNullException	Negative
TC-25	Defensive / Edge	Empty string returns empty string	Method available; valid input string	""	""	Edge Case
TC-26	Defensive / Edge	Only # returns empty string	Method available; valid input string	#	""	Edge Case
TC-27	Defensive / Edge	Only spaces and # returns empty string	Method available; valid input string	" #"	""	Edge Case

3. Test Type Legend

Positive: Input is valid; method should return the expected output

Negative: Input causes an expected exception (e.g. ArgumentNullException)

Edge Case: Boundary or unusual conditions — empty string, only backspaces, wrap-around